7.1 Angles of Polygons

Geometry 7.1 · Student sheet

Name _____ Period ____ Date _____

I can…

- find the sum of the interior angle measures of a polygon.
- use interior and exterior angle sums to find unknown angle measures.
- find each interior and exterior angle of a regular polygon.

Vocabulary · fill in the missing word

Diagonal — a segment that joins two _____ vertices of a polygon

Exterior angle — an angle formed by one side and the _____ of the next side

Equilateral / equiangular polygon — all sides _____ / all angles congruent

Regular polygon — a convex polygon that is both _____ and equiangular

Part 1 · Discovery: triangles in polygons

Polygon names (keep this list for the whole lesson)

Sides	Name	Sides	Name
3	triangle	8	octagon
4	quadrilateral	9	nonagon
5	pentagon	10	decagon
6	hexagon	12	dodecagon
7	heptagon	n	n-gon

Draw a convex quadrilateral, pentagon, hexagon, and heptagon. From ONE vertex, draw every diagonal. Count the triangles.

Polygon	Number of sides	Triangles from one vertex	Sum of the interior angles
Triangle	3		
Quadrilateral	4		
Pentagon	5		
Hexagon	6		
Heptagon	7		
n-gon	n		

Exterior angles. Draw a polygon, extend one side at each vertex, cut out the exterior angles, and put them together around a point. What do they make?

Part 2 · Notes and examples (fill in each note, then work the example)

Polygon Interior Angles Theorem: The sum of the interior angle measures of a convex n-gon is _____.

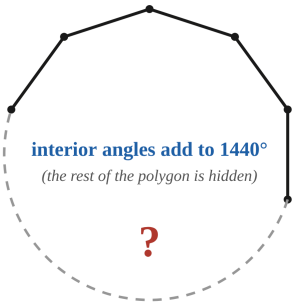
Corollary: The interior angle measures of a quadrilateral add to _____°.

Example 1

Find the sum of the interior angle measures of a convex 12-gon.

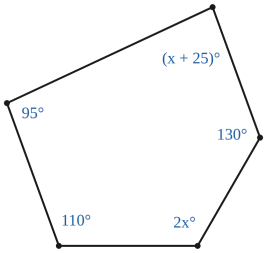
Example 2

The sum of the interior angle measures of a convex polygon is 1440°. Classify the polygon by its number of sides.



Example 3

A pentagon has interior angles of 110°, 2x°, 130°, (x + 25)°, and 95°. Find x and the two unknown angles.



Polygon Exterior Angles Theorem: The exterior angles of a convex polygon, one at each vertex, add to _____°.

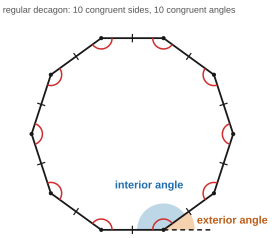
Example 4

A quadrilateral's exterior angles (one at each vertex) are x°, 2x°, 89°, and 67°. Find x.

Regular n-gon: each interior angle = _____ each exterior angle = _____

Example 5

Find the measure of each interior angle and each exterior angle of a regular decagon.



Part 3 · Partner practice

Partner 1

Find the sum of the interior angle measures of a convex 15-gon.

Partner 2

Each exterior angle of a regular polygon measures 24°. How many sides does it have?